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Machine learning determination of atomic dynamics at grain boundaries

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In polycrystalline materials, grain boundaries are sites of enhanced atomic motion, but the complexity of the atomic structures within a grain boundary network makes it difficult to link the structure and atomic dynamics. Here, we use a machine learning technique to establish a connection between local structure and dynamics of these materials. Following previous work on bulk glassy materials, we define a purely structural quantity (softness) that captures the propensity of an atom to rearrange. This approach correctly identifies crystalline regions, stacking faults, and twin boundaries as having low likelihood of atomic rearrangements while finding a large variability within high-energy grain boundaries. As has been found in glasses, the probability that atoms of a given softness will rearrange is nearly Arrhenius. This indicates a well-defined energy barrier as well as a well-defined prefactor for the Arrhenius form for atoms of a given softness. The decrease in the prefactor for low-softness atoms indicates that variations in entropy exhibit a dominant influence on the atomic dynamics in grain boundaries.

grain boundary diffusion | atomic plasticity | machine learning | materials science | nanocrystalline

Atomic rearrangements, in which atoms overcome energy barriers to change neighbors, underpin the dynamics of grain boundaries (GBs) in polycrystalline materials that are ultimately responsible for such phenomena as grain growth (GB migration), GB diffusion, GB sliding, emission/absorption of lattice dislocations, and point defect sink behavior. Assessing the atomic-scale dynamics of GBs from their structure in realistic polycrystalline materials is inherently complicated. This complexity is associated with both the highly degenerate nature of GB structures (1–3) as well as the interconnected nature of the GB network within polycrystalline microstructures (e.g., associated with GB junctions, compatibility, etc.). The ensemble of possible configurations produces a nearly continuous spectrum of energies and a correspondingly large set of local structures (3). The atomic structure at the GB exhibits some order imparted by the contiguous grains but also becomes trapped in metastable minima in a complex energy landscape reminiscent of that of a glass (3–5). This suggests that methods appropriate for glass dynamics may be fruitful for understanding the dynamics internal to GBs in polycrystals.

Previously, it has been shown that low-frequency quasilocalized vibrational modes can be used to identify atoms that are likely to rearrange in both glasses (6, 7) and GBs (8). Here, we push the analogy between GBs and glasses further by using a machine learning analysis, originally developed to study atomic rearrangements in glasses, to characterize the local atomic rearrangements in molecular dynamics simulations of polycrystalline solids. The rearrangements occur as atoms thermally fluctuate over local barriers between metastable sites. On long timescales, asymmetries in these dynamics give rise to kinetic phenomena, such as GB migration, creep, and defect emission, but here we consider the individual atomic rearrangements within the GB.

Because defect microstructures are commonly spatially extended, it is not clear that atomic rearrangements can be characterized in terms of only local structural information at the atomic scale. Here, we show that structural information within a few atomic diameters is indeed sufficient to predict these atomic-scale rearrangements and characterize particles in terms of a single continuous scalar variable called "softness," which captures the relevant properties of the local atomic environment. Remarkably, we find that particles of a given softness are characterized by a well-defined common energy barrier to rearrangements in polycrystals, just as was previously discovered for glassy liquids (9). Thus, our results translate into a spatial map of the energy barriers to rearrangements. Our findings suggest that it is possible to characterize much GB dynamical behavior using only local, atomic-scale structural information.

Methods

Large-scale molecular dynamics simulations (10) of nanocrystalline aluminum are performed as in ref. 11, producing a network of interacting GBs. The geometry is initialized using a random (Poisson point process) Voronoi tesselation and curvature–flow grain growth algorithm (12, 13). Atomic interactions model aluminum using the embedded atom method

Significance

A machine learning method is used to analyze the atomic structures that rearrange within the grain boundaries of polycrystals. The method readily separates the atomic structures into those that rarely rearrange and those that often rearrange. The likelihood of an atom rearranging under a thermal fluctuation is correlated with free volume and potential energy but is not entirely attributable to those quantities. A machine-learned quantity allows estimation of the energy barrier to rearrangement for particular atoms. The grain boundary atoms that rearrange most have more possible rearrangement trajectories rather than much-reduced energy barriers, as in bulk glasses. The work suggests that polycrystal plasticity can be studied in part from the local atomic structural environments without traditional classification of microstructure.

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Data deposition: Data, simulation software, and input files have been deposited in GitHub, github.com/SimonsGlass/GBAnalysisData.

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(14). The grains are face-centered cubic (FCC) with nominal nearest neighbor distance d of approximately 2.8 Å and a melting temperature of $T_m = 926$ K. The system of 4 million atoms is simulated at a fixed temperature, T, using a Nosé–Hoover thermostat. The atomic positions are averaged over 0.1-ps intervals to reduce the highest-frequency atomic vibrations.

The microstructure of the simulated nanocrystalline aluminum can be visualized using an adaptive common neighbor analysis (CNA) (15) (Fig. 1*A*) or Voronoi topology (VoroTop) analysis (12, 16) (Fig. 1*B*) to distinguish grains from GBs. Both CNA and VoroTop assess local crystalline structure around an atom using the relative positions of the atom's neighbors. CNA defines neighbors as those atoms within a cutoff distance somewhat larger than d and compares which neighbors are also neighbors of each other. Lists of shared neighbors are then compared with those generated from ideal crystalline lattices. In contrast, VoroTop does not impose a cutoff distance but analyzes the atomic Voronoi cell structure to determine similarity to cells from perturbed crystalline lattices. This method more readily identifies crystalline environments, recognizing 89% of atoms in the simulation in Fig. 1 as FCC compared with 78% by CNA. The VoroTop analysis is also much more robust against thermal fluctuations, so that the identified GB regions do not grow significantly with temperature up to 0.8 T_m.

Our aim is to characterize the propensity for an atom to rearrange. We follow a procedure designed for glassy systems (9, 17–19) that uses a support vector machine (SVM; a supervised machine learning technique) (20) to identify correlations between the local structure around each atom and its rearrangements. Complete details are given in *SI Appendix*. A descriptor, or fingerprint, of the local structure around each atom is calculated by evaluating a set of so-called structure functions (SFs) (21). SFs are functions of the relative positions of the atoms out to a cutoff distance $d_0 = 5.0$ Å from a central atom, about twice the interatomic spacing. We use two types of SFs to ensure that different atomic environments lead to different fingerprints (21, 22). The first type depends on the radial distribution of neighbors from the central atom, whereas the second type encodes the distribution of triangles of varying angles formed by the central atom with pairs of neighbors. (The complete functions are given in *SI Appendix*.)

Next, we identify which atoms rearrange in the course of the dynamical simulation. For this, we use a standard atom-based quantity, $p_{hop}(t)$ (23, 24); $p_{hop}(t)$ becomes large when the atom moves a long distance on the time scale of atomic vibrations (*SI Appendix*). Examination of Fig. 1*C* shows that p_{hop} is only large at the GBs (marked as red or blue atoms in Fig. 1 *A* and *B*). Rearrangements are defined as events that exceed the threshold $p_{hop} > p_c = 1.0$ Å^2. There is also considerable variation within and between GBs; although 22% of the atoms are in disordered environments in our microstructure according to CNA, at any instant fewer than 0.06% of atoms are rearranging, and only 0.5% of the atoms have $p_{hop} > 0.5$ Å^2.

To correlate rearrangements to atomic environments, we construct a training set for the supervised machine learning algorithm. We choose atoms to put into the training set based on whether they arrange within a 200-fs window (we label these $y_i = 1$) or do not rearrange over a much longer time (1.8-ps) period (we label these $y_i = -1$).

The local structural environment around any atom is described as a point in a high-dimensional space in which each orthogonal axis corresponds to possible values of a different SF. The SVM identifies the hyperplane in this space that best separates the two groups of atoms in the training set (generalized linear regression). We find that 96% of atoms with $y_i = 1$ fall on what we define as the "positive" side of the hyperplane and that 90% of the atoms with $y_i = -1$ fall on the other ("negative") side of the hyperplane. This justifies using a hyperplane rather than a more general surface to separate the two groups.

Results

Linking Structure and Dynamics. For any atom, the point characterizing its local structural environment can then be compared with the hyperplane in SF space. The signed normal distance from the point to the hyperplane defines the value of the softness, S (9). Fig. 1*D* shows the system with each atom colored according to its softness. Atoms with large positive softness (red in Fig. 1*D*) tend to lie in GBs, while those with large negative softness (blue in Fig. 1*D*) lie within the grain interiors as expected. In *SI Appendix*, we find that softness is partially correlated with other structural quantities, such as free atomic volume, but that softness is considerably more predictive of whether atoms rearrange.

We note several interesting features in Fig. 1 associated with the considerable variation in softness along the GBs in the system. Comparing Fig. 1*A* with Fig. 1*D*, we see that the GB above the label (1) has small softness relative to the other GBs in the system. Examination of the structure of that GB shows that it is a large-angle twist GB lying along a close-packed {111} plane. Such a GB provides only very localized distortions to the crystal structure of the grains, and few atomic rearrangements occur there.

At label (2) in Fig. 1, the viewing plane is nearly coplanar with the GB, showing both structural and dynamic heterogeneity within the GB. The position in the microstructure labeled (3) in Fig. 1 shows the intersection of a coherent twin boundary with a more general GB. The twin boundary is no softer than the grain interior. Its intersection with the GB, however, changes the GB character (misorientation) and the resulting softness such that the segment of the GB above the intersection with the twin boundary is significantly softer than the segment of the GB just below the intersection. A lattice vacancy is seen near label (4) in Fig. 1; it is softer than the surrounding lattice but not as soft as many of the GB sites. This is consistent with earlier work that analyzed vibrational modes and found that vacancies were more mechanically stable than GBs (25). Indeed, we find that vacancies have softness $S \approx 0$.

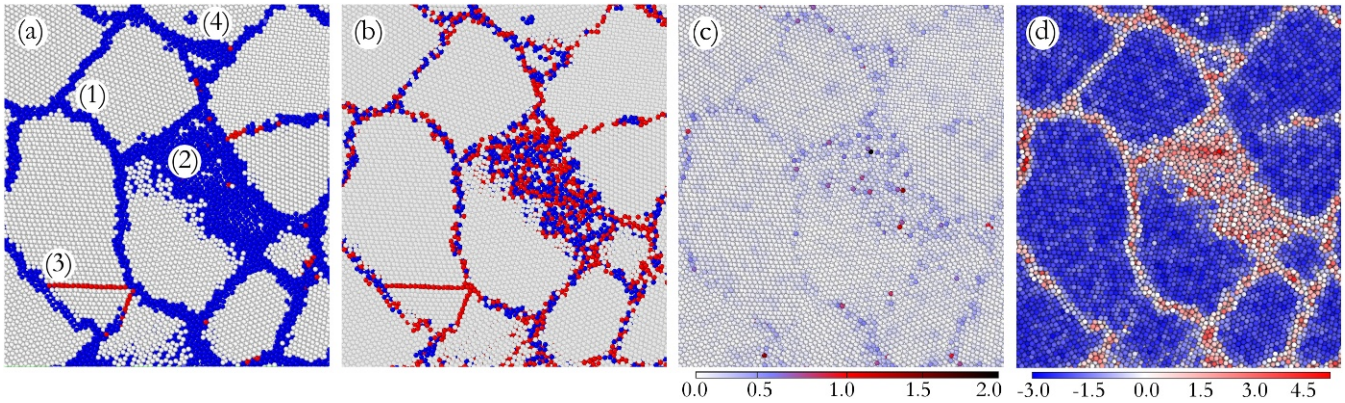

Fig. 1. A small region of a cross-section through the 3D polycrystalline aluminum microstructure at $T = 463$ K as visualized by (*A*) CNA and (*B*) VoroTop (12). Grain interiors are locally FCC (white atoms), while stacking faults and twin boundaries are visible as locally hexagonal close-packed (HCP) structures (red atoms); atoms within general GBs are neither FCC nor HCP (blue atoms). Positions are averaged over a 0.1-ps window. (*C*) The same atoms colored by their instantaneous value of p_{hop} (Å^2) as indicated by the color bar. (*D*) The same atoms colored according to their softness as indicated by the color bar.

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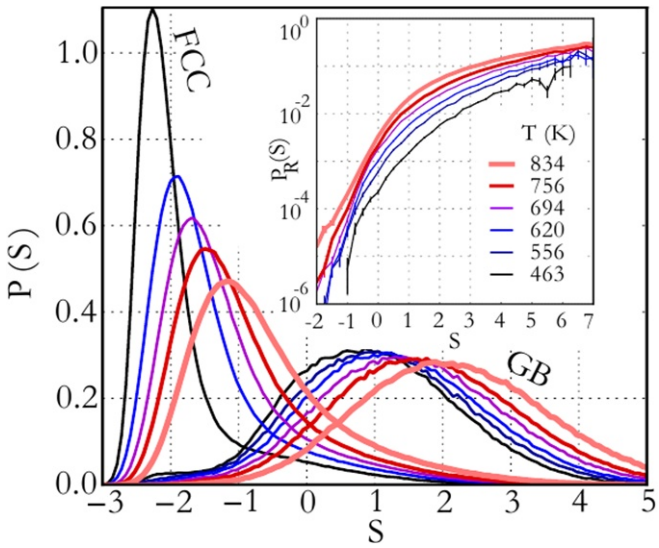

Fig. 2. The distribution of softness, S, within grain interiors and at GBs. The interiors are distinguished from GBs on the basis of their VoroTops. About 89% of atoms are in an FCC environment. (*Inset*) The probability to rearrange, P_R, rises monotonically with softness, S, over several orders of magnitude for all temperatures.

Fig. 2 puts these results into context. Here, we decompose the distribution of softness from the polycrystalline system into contributions from the grain interiors (i.e., FCC regions as determined by the VoroTop analysis) and all other regions. The least soft atoms are FCC, while the softest are associated with defects. Softness is more homogeneous in the grain interiors than in the regions that VoroTop defines as disordered (mostly GBs). The softest atoms tend to lie near the center of the GBs. In both regions, the distribution shifts slightly with temperature, reflecting a shift in the distribution of local atomic configurations due to thermal distortions of the crystalline grains as well as an increase in the effective GB thickness.

Although we used only a binary classification to define the hyperplane, the magnitude of the softness is predictive of dynamics. We establish this by studying the probability that an atom rearranges, $P_R(S)$, defined as the time average of the fraction of atoms in the system with a given S with $p_{hop} > p_c$. Fig. 2, *Inset* shows that the fraction of atoms with a given S that will rearrange, $P_R(S)$, is a strongly increasing function of softness S. $P_R(S)$ increases monotonically with S, and $S = 1$ atoms are more than 100 times more likely to rearrange than $S = -1$ atoms. At large S, $P_R(S)$ saturates, as the fraction of hopping atoms cannot exceed one.

Extracting the Energy Barriers of Atomic Rearrangements. In earlier work on glassy systems, it was discovered that the probability that an atom rearranges, $P_R(S)$, is Arrhenius for each value of S (9, 19, 26), implying a well-defined energy barrier associated with rearrangements, $\Delta E(S)$. We, therefore, study the temperature dependence of $P_R(S)$. Fig. 3 shows that the temperature dependence of $P_R(S)$ is indeed well described as Arrhenius at temperatures $T < 0.8\ T_m$, showing that softness reflects the energy barrier for an atom to rearrange.

The Arrhenius form implies that the probability to rearrange can be written

$$P_R(S) = e^{\Sigma(S)} e^{-\Delta E(S)/k_B T}. \quad [1]$$

Fig. 4 indicates that the effective energy barriers decrease slightly with softness, changing by less than a factor of two over the observed range of softness. The magnitude of the energy barrier is consistent with nudged elastic band calculations of structural transitions between specific metastable states in bicrystals using the same interatomic potential used here (27). The dominant energy barriers, of order 100 meV, are large compared with applied elastic stress (the yield stress of Al is ~ 10 MPa or 1 meV). This shows that these atomic rearrangements will be largely thermally activated and that applied stresses have only little effect, and it suggests that asymmetries in transition rates underlie stress-driven microstructure evolution.

Changing softness has much more of an effect on the prefactor in the Arrhenius relationship, Σ, than it does on the barrier ΔE. For a thermally activated process, Σ can be viewed as a generalized attempt frequency or alternatively, as an entropic contribution to the free energy barrier. The increase of Σ with softness suggests that soft atoms have more directions for rearrangement or more paths that can take them to the transition state. This is in contrast to bulk glasses in which it was found that both ΔE and Σ decreased with increasing S. We, therefore, see that the reason that softer atoms are more likely to rearrange in GBs is that they are the ones with slightly lower-energy barriers and substantially increased Σ.

We consider the origin of this difference from bulk glasses. In polycrystals and unlike in bulk glasses, there are crystalline grains that restrict the rearrangements. Atoms in the crystalline grain mostly cannot participate in rearrangements due to the large potential energy barrier to move them significantly. With fewer atoms able to participate, the number of possible rearrangement trajectories (and Σ) is reduced. *SI Appendix* shows explicitly that low-softness ($S \approx 0$) atoms are also in the most crystal-like neighborhoods as indicated by Voronoi volume, radial symmetry, and potential energy. These crystalline neighborhoods evidently restrict Σ sufficiently to have a large impact on the GB dynamics, leading to low-Σ, low-S atoms.

The best fit lines in Fig. 3 indicate no common intersection point. This implies that there is no one temperature at which

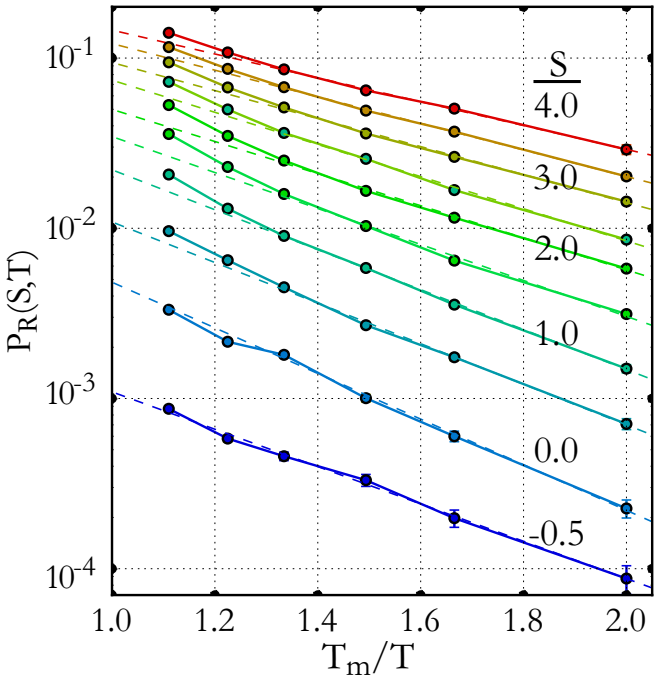

Fig. 3. The probability of rearrangement exhibits Arrhenius behavior at temperatures below 0.80 T_m. Fits to the lower four temperatures are shown as dashed lines.

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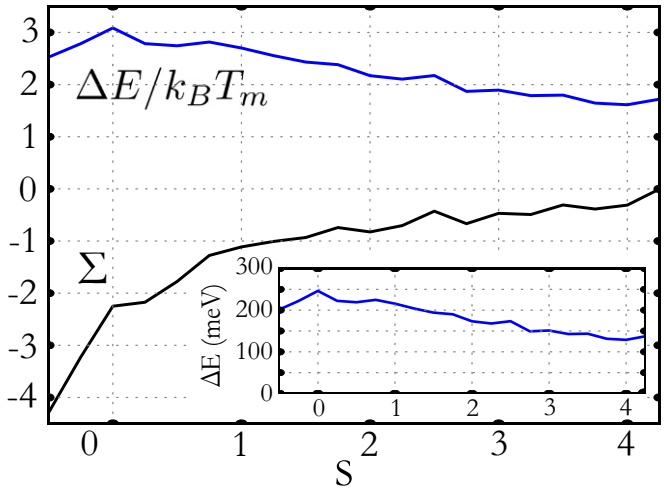

Fig. 4. The extracted energy barrier scaled by the thermal energy at the melting temperature $\Delta E/k_BT_m$ (blue) and Σ (black) extracted from the rearrangement probability (Eq. **1**). (*Inset*) ΔE vs. S. The energy barrier varies slightly with S, while Σ increases strongly with S.

rearrangement dynamics become independent of softness or local structure. This contrasts with the behavior of glassy systems, which display a common intersection point (9) at what is known as the onset temperature. This temperature marks the onset of features associated with supercooled liquids, such as nonexponential relaxation, a non-Arrhenius dependence of the relaxation time on temperature and kinetic heterogeneities. The fact that there is no indication of an onset temperature for this system indicates that GBs, despite exhibiting some glass-like properties (4), also exhibit behavior that is very unlike glasses.

Analysis of SVM. We next analyze which SFs are most responsible for the softness of the atom. Recall that the structural quantities that we use to characterize the local environment of each atom fall into two classes: radial SFs depend on only the radial distribution of atoms from the central atom, and angular SFs additionally depend on the angles formed with pairs of neighboring atoms around the central atom. In bulk glasses, angular SFs were unimportant compared with the radial SFs, and in fact, softness was almost entirely attributable to the number of neighboring atoms at the distances of the first peak and valley of the pair correlation function of the material, $g(r)$ (9). In GBs, one may expect increased importance of angular SFs, since they are sensitive to the relative orientations of the lattices (i.e., the crystallography of the GB).

To connect with studies on bulk glasses, we exclude crystalline atoms from the training set using CNA, although we find that this does not significantly change the relative importance of most SFs. Excluding crystalline atoms does, however, decrease the fraction of training set atoms that are accurately classified from 93 to 79%, since now the SVM discriminates solely between disordered atoms.

We use recursive feature elimination (RFE) to determine the important features. In RFE, the SVM is first trained using all SFs. Then, the SF that contributes the least weight to the hyperplane is identified and eliminated. Training is repeated, iteratively identifying and eliminating the least important SF and simplifying the fingerprint while attempting to retain the highest accuracy.

Fig. 5 shows the decrease in accuracy as the number M of SFs decreases. The color of the symbol indicates the class of the least significant SF, which then becomes eliminated. All radial functions are eliminated quickly (red hollow circles in Fig. 5), decreasing accuracy f_a only slightly to 77%. Of 72 total features, the last 47 features to be eliminated (and therefore, the top 47 most important quantities) are all angular functions. This is in stark contrast to bulk glasses, where all of the angular functions could be eliminated entirely with less than a 2% cost in accuracy (9); 14 angular functions are sufficient to retain 77% accuracy, and accuracy remains near 69% using the sole SF identified as most important. The increased importance of angular information here may seem natural given the role of the degree and relative orientation of the nearby crystallinity in allowing rearrangements. The contribution of each specific SF to softness is quantified in *SI Appendix*. The properties of the specific structures that lead to rearrangements should be explored further.

Discussion

In summary, we have used machine learning to introduce a structural quantity (softness) that is strongly correlated with the dynamics of atomic rearrangements in GBs. Correlations between local structure and rearrangements are so strong in these nanocrystalline metals that over 96% of the observed atomic rearrangements correspond to atoms for which $S > 0$. The only information that enters the machine learning is provided by the training set, which is chosen to represent atomic environments just about to rearrange and those that do not rearrange. A binary classification on this training set yields a remarkably rich lode of information. Softness can distinguish certain GBs, where rearrangements are imminent, from those that are markedly static, and it can reveal differences even within a given GB. It also gives information about the energy landscape of the system, providing both the typical energy barriers and the attempt frequencies for atoms to rearrange—based solely on the their local structural environments. Such information is computationally expensive via existing methods. Interestingly, we show (*SI Appendix*) that, while softness correlates to some degree with other structural parameters that identify defects in crystals, it is remarkably superior in identifying which defects/local environments are active in GB dynamics.

Our conclusion that the variation of entropy plays a central role in the atomic rearrangements at GBs is reminiscent of prior observations about nanoconfined viscous fluids. Specifically, in refs. 28 and 29, it was found that the total thermodynamic entropy of the confined fluid [specifically, the excess entropy (the entropy above that of an ideal gas at the same density and temperature)] closely reflects the rate of structural relaxation. The atoms in GBs are somewhat analogous to a confined fluid, noting that the complicated confinement in GBs is due to structured

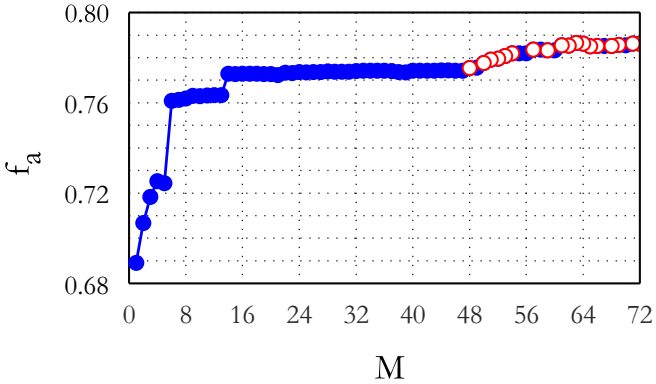

Fig. 5. RFE determination of the most important SFs. The fraction of training set atoms that are accurately classified, f_a, decreases as SFs are eliminated. The color of the symbol shows whether the least important SF—which then gets eliminated—is a radial SF (red hollow circle) or an angular SF (blue solid circle).

surfaces of crystalline grains that can themselves rearrange and the configurations are not in equilibrium. Our approach attempts to infer the local contributions to the entropy that are involved in the rearrangements. The observation from Fig. 4 that structures that tend to rearrange more frequently are associated with higher entropy constitutes a local version of the statement that the entropy underlies the relaxations. However, the characteristic local energy barrier additionally emerges from our analysis of atomic dynamics.

Our approach shows that focusing on the dynamics at the atomic scale can serve as a viable alternative to the classification of environments based on the rich and complex zoology of crystallographically allowed defects. This approach, successfully applied originally to glasses (9, 19), suggests that it is possible to construct a single framework to describe atomic-scale dynamics in systems with varying degrees of order/disorder. At the very least, this approach is complementary to existing methods that strive to relate dynamic materials phenomena to the underlying structure of their hosts. More optimistically, we note that the larger-scale dynamics are dictated by the interplay of softness with atomic rearrangements—while softness predicts the propensity to rearrange, a rearrangement alters local structure and hence, softness. Understanding this interplay is a step toward constructing a theory of plasticity that has the potential to span the entire gamut of materials from crystalline to glassy.

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